

Noah's Ark in a Warming World: Climate Change, Biodiversity Loss, and Public Adaptation Costs in the United States

Frances C. Moore, Arianna Stokes, Marc N. Conte, Xiaoli Dong

Abstract: Climate change poses a growing threat to biodiversity, but the welfare consequences of these changes are not well understood. Here we analyze data on the US Endangered Species Act and project increases in species listing and spending due to climate change. We show that higher endangerment is strongly associated with the probability of listing but also find a large bias toward vertebrate species for both listing and spending. Unmitigated warming would cause the listing of an additional 690 species and committed spending of \$21 billion by 2100. Several thousand more species would be critically imperiled by climate change but remain unlisted. Finally, we compare ESA spending with estimates of willingness to pay for conservation of 36 listed species. Aggregate WTP is larger than ESA spending for the vast majority of species even using conservative assumptions and typically one to two orders of magnitude larger than direct ESA spending using less restrictive assumptions.

JEL Codes: H41, Q51, Q54, Q57, Q58

Keywords: endangered species, climate change, biodiversity, adaptation

HUMANS COEXIST ON EARTH with immense, but still not fully understood, biological diversity—likely the richest assemblage of life to ever exist on the planet (Benton

Frances C. Moore (corresponding author) is at the Department of Environmental Science and Policy, University of California Davis, 2140 Wickson Hall, One Shields Ave, Davis, CA 95616 USA (fmoore@ucdavis.edu). Arianna Stokes is at the Department of Environmental Science and Policy, University of California Davis. Marc N. Conte is at the Department of Economics, Fordham University. Xiaoli Dong is at the Department of Environmental Science and Policy, University of California Davis. We thank Rudy Huezo and Abigail Warner for excellent research assistance. Ideas contained here benefited from discussions with Mark Schwartz. The work was funded by NSF CNH2-S: Understanding the Coupling between Climate Policy and Ecosystem Change (award 1924378).

Dataverse data: <https://doi.org/10.7910/DVN/2FDEFO>

Received August 5, 2020; Accepted July 19, 2021; Published online July 5, 2022.

Journal of the Association of Environmental and Resource Economists, volume 9, number 5, September 2022.
© 2022 The Association of Environmental and Resource Economists. All rights reserved. Published by The University of Chicago Press for The Association of Environmental and Resource Economists.
<https://doi.org/10.1086/716662>

1994). Human influence has, however, taken a toll on this diversity. Human beings appropriate an estimated 25% of the Earth's net primary productivity and 35% of its land area to support just a few dozen domesticated crop and livestock species, reducing the richness and abundance of natural systems (Krausmann et al. 2013; Newbold et al. 2015; FAO 2016). Almost 200 vertebrate species have gone extinct since 1900, a rate at least eight times the background rate inferred from the fossil record (Ceballos et al. 2015).

Anthropogenic climate change poses a further threat to global biodiversity through a number of pathways—the climatological niche may become too small to support a viable population; limits to dispersal mean species may not be able to move with shifting climate envelopes; phenological changes could disrupt food webs, particularly for migratory species; and widespread tree die-offs and wildfires linked to climate change could directly destroy habitat (Bellard et al. 2012). Climate change has already been implicated in the extinction of at least one species (Pounds et al. 1999; Thomas et al. 2004), and projections for the future are that unmitigated climate change over the twenty-first century will threaten somewhere between 10% and 15% of species with extinction (Urban 2015). The US Fish and Wildlife Service (USFWS) has already listed the polar bear and has proposed listing the North American wolverine, primarily due to the risk of habitat loss from climate change (Blumm and Marienfeld 2014). Increasing extinction risk has implications for current and future generations, but this important climate change cost is not well understood.

Estimating the costs of increasing extinction risk is challenging. Accounting for the value of a species' simple existence requires application of stated-preference nonmarket valuation methods. The most notable of these methods, contingent valuation, has been the subject of ongoing debate for nearly 30 years (e.g., Hausman 2012; Kling et al. 2012; Haab et al. 2013). This continued debate has left existence values often sidelined in the evaluation of conservation policies, despite the development of alternative choice experiment methods that address many of the original methodological concerns (Adamowicz et al. 1994; Hanley et al. 1998).

Despite large gaps in the data that economists would normally use in policy evaluation, it is an empirical fact that many governments around the world, particularly those in richer countries, devote resources to preventing extinctions (Waldron et al. 2013). Although separating consumptive use from nonconsumptive and nonuse values may be challenging, most wealthy countries have policies, laws, and spending that aim explicitly to prevent species extinctions, irrespective of any consumption value those species do or do not provide for people.

This fact raises both positive and normative questions for environmental economics. First, given that biodiversity preservation is a policy goal for many governments, economics can provide insight into the most cost-effective means of meeting those goals: how should a limited budget be prioritized so as to protect the largest diversity of species? Second, patterns of spending on biodiversity may reveal a signal, albeit a noisy signal filtered

through a complex political process, of public preferences over different and competing priorities for species protection. Martin Weitzman tackled both these questions in an extensive research program over the 1990s.

In a series of papers, Weitzman developed a mathematical description of diversity based on pairwise dissimilarity, formally connecting it to the branching tree structure often used to describe evolutionary relationships and the genetic information content of extant species (Weitzman 1992, 1998). Using data on crane species, he provided an illustration of how this approach could be applied to inform conservation priorities, highlighting the complex interactions between evolutionary relationships, probability of survival, and the effectiveness of conservation spending that determine the optimal allocation of resources (Weitzman 1993). In the well-known Noah's Ark problem, Weitzman (1998) advanced the theory of conservation spending further, developing a simple and robust ranking for cost-effective spending that depends on evolutionary distinctiveness, utility value to humans, and the effectiveness of spending at increasing survival probability.

In two other papers, Weitzman, together with Andrew Metrick, analyzed what was then still new data on federal and state spending under the US Endangered Species Act (ESA) (Metrick and Weitzman 1996, 1998). In contrast to the line of work just described, these papers are primarily descriptive, an attempt to empirically distinguish the priorities implied by conservation decisions amid the "shopping list of objectives" typically used to justify it: "Decisions about endangered species reflect the values, perceptions, and contradictions of the society that makes them. Thus . . . this paper addresses some very general issues about humankind's relation to nature and about our choices when confronted by competing and often unquantifiable objectives" (Metrick and Weitzman 1996, 1). In these papers, Metrick and Weitzman developed several proxies of the variables that appear in the optimal allocation results in the Noah's Ark problem, such as distinctiveness (measured as a species being the only member of its genus), utility value (measured using taxonomic group and body size), and degree of threat (measured using a scientific evaluation by the Nature Conservancy). They found evidence that both scientific considerations (i.e., degree of threat and biological distinctiveness) and variables measuring the utility value to humans play a role in both the listing and spending decisions.

Other papers have since examined the effect of ESA spending on species recovery (Kerkvliet and Langpap 2007), the optimal allocation of ESA spending (Langpap and Kerkvliet 2010), and the costs of other aspects of endangered species protection such as controls over federal land use or restrictions on private development (Ando 2001; Langpap et al. 2018) and the political economy of lobbying over listing decisions (Ando 2003). However, few recent papers have revisited Metrick and Weitzman's original question of what social priorities seem to govern either the listing of species or the allocation of resources between species (Dawson and Shogren [2001] and Kerkvliet and Langpap [2007] being the exceptions).

Here we develop models of ESA listing and spending and combine them with projections of increasing extinction risk under climate change to provide a rough estimate of the

change in the number of listed species, spending under the ESA, and species that will be imperiled by climate change but remain unlisted in order to bound the costs associated with climate-change driven biodiversity loss in the United States. These costs can be broken down into four parts, of which we explicitly quantify only the first:

1. Direct spending on the conservation of protected species under the ESA. These could include conservation activities such as research, census, habitat maintenance or transplantation as well as land and habitat acquisition. These are the costs considered in this paper.
2. Direct spending on conservation is a form of public adaptation. Because public funds must be raised from distortionary taxes, each dollar of public spending produces additional effects that lead the total welfare cost to be greater than \$1. Barrage (2020) has demonstrated the importance of these effects for the welfare costs of climate change and suggests that they increase the costs of direct public spending by 4%–53%, depending on the distortionary effect of the revenue-raising mechanism.
3. ESA listing entails protections that place limits on the use and development of private land. These restrictions, discussed more fully in section 3, entail opportunity costs that have not been systematically quantified for all listed species. It is likely that these would increase with the number of listed species, but they are not quantified here.
4. Our results suggest that only a small fraction of species imperiled by climate change will be listed, meaning that climate change will likely cause extinction of some species. The lost existence value from these extinctions cannot be quantified in this paper. Section 4, however, provides a comparison between direct spending on particular endangered species and willingness-to-pay (WTP) estimates derived from stated preference studies, providing some indication of the order of magnitude of these costs relative to direct spending, at least for this small subset of species.

Beyond our assessment of climate-change-driven costs, the paper extends the original analysis by Metrick and Weitzman in several ways. First, the quantity of spending data has increased substantially since the early 1990s. This results both from a longer (24-year) time series and a much larger set of listed species. This much longer data set allows us to avoid relying on the pooled cross-sectional models used in Metrick and Weitzman (1996, 1998) and to estimate a model with species fixed effect that controls for unobserved variation between species, comparing it to a random-effects model that allows estimates of important time-invariant characteristics. In addition, Metrick and Weitzman's (1996, 1998) analysis was limited only to vertebrate species, meaning it is not clear how their findings might generalize to the far more numerous set of nonvertebrate species. Our analysis

also includes plants and invertebrates, increasing the number of species considered from 511 in Metrick and Weitzman (1996) to over 64,500.

Next, advances in genetics mean that it is now possible to directly quantify the evolutionary distinctiveness of a large number of species, a measure of the genetic distance of a particular species from its nearest relatives. This concept figures prominently in Weitzman's theoretical results on optimal conservation spending (Weitzman 1992, 1993) but is proxied only very imperfectly by taxonomic measures that have been previously used to assess conservation spending. We include recently available measures of the species phylogenetic distinctiveness as a variable explaining listing decision to assess the degree to which this has been incorporated into conservation policy.

Finally, the paper begins to address the question of the benefits of ESA listing through a comparison of the WTP for conservation of endangered species with direct ESA spending on those species. Interpreting ESA listing as an adaptation to climate change implies that the benefits exceed the costs. Although comprehensive benefit-cost analysis of ESA listings is not possible here, this review of values reported in the stated preference literature begins to bound the benefit-cost ratios associated with listing.

The paper proceeds as follows: the next section introduces the data sources used. Section 2 presents statistical modeling results for two models, the listing status and the spending decision. Section 3 combines findings from these regressions with prior estimates from the ecological literature of the threat climate change poses to North American biodiversity to estimate the increase in species listing and spending with climate change. Section 4 provides a review of WTP estimates for 36 listed species and compares these values with the observed ESA spending. Section 5 concludes.

1. DATA SOURCES

1.1. Listing and Spending

Our primary dependent variables of interest are (1) listing a species as threatened or endangered and (2) the level of spending received by that species, conditional on listing. These two factors will jointly determine how the increased risk of extinction posed by climate change translates into higher fiscal burdens under the ESA. Although the ESA requires the listing of species at risk of extinction, capacity and budgetary constraints mean that the process is slow and there is a backlog of species awaiting consideration (Alexander 2010). In some cases, USFWS will make a "warranted but precluded" designation, explicitly acknowledging that the biological threat of a species requires listing but denying listing because other species are a higher priority (Alexander 2010). Therefore, it seems likely that which species do or do not receive listing protection may be influenced both by scientific assessments of endangerment as well as other social, political, and economic factors that determine priority.

US Fish and Wildlife Service (USFWS) funding, which constitutes about 15% of the reported expenditures, is distributed among the eight USFWS administrative regions based on the total number of species assigned to each region and the estimated recovery

costs of those species. Regional offices then distribute funds to field offices, a process that may be heavily influenced by field offices' "long-standing arrangements to work with partners to recover specific species" (GAO 2005). Field offices use funds to implement recovery plans of species under the office's jurisdiction, based on priority rankings (described further below) and partnerships with outside organizations. Therefore, like the listing decision, the spending decision also likely reflects both scientific factors such as the level of endangerment or potential for recovery and the utility value (i.e., popularity) of particular species.

Species-specific federal and state spending on endangered species recovery is reported every year by the USFWS and the National Marine Fisheries Service (NMFS). The NMFS oversees a small set of coastal and marine species—they constitute 46 out of 1,613 listed species in our listing regression but are dropped from our spending analysis because of the lack of covariate data, particularly on range area. Expenditures, which includes both spending on recovery efforts and land acquisition, were extracted from these reports for the period 1993–2016. Expenditures were adjusted to real 2017 dollars using the Bureau of Labor Statistics' Consumer Price Index. Entities eligible for protection by the ESA include species, subspecies, and distinct population segments (DPS). When expenditures were reported separately for distinct population segments of the same species, expenditures were aggregated to the species or (where appropriate) subspecies level. This is necessary since ESA spending and recovery data for DPS were not reported separately for many listings prior to 2003 and other covariates (such as conservation status and n-gram data, described more fully below) are not available at the DPS level.

The species-specific spending documented in these reports includes all spending that may be reasonably attributed to an individual species. Reported expenditures capture a range of conservation activities such as research, census, habitat maintenance, propagation, live trapping, and translocation, as well as land acquisition, employee salaries, listing, consultation, and law enforcement costs where these can be attributed to individual species (USFWS 2016). Reports include spending by both federal and state agencies, with the vast majority (92%) attributable to federal agencies. Spending is disaggregated by land acquisition versus other conservation costs, with land acquisitions constituting 20% of total expenditures. General operational expenditures that cannot be attributed to a particular species are not reported and are not included in this analysis.

Table 1 and figure 1A give some summary figures from the spending data. Figure 1A shows median annual spending by taxon across the 24 years in the data set. Large differences across groups of species are apparent in the raw data, in particular an order-of-magnitude difference between the vertebrate species (birds, mammals, reptiles, amphibians, and fish) and the plants, invertebrates, and fungi. Table 1 shows the 25 species that receive the most spending and highlights the skewed distribution of spending among species, with a large emphasis on salmonid species, which are unusual among endangered species in having a large recreational and commercial value.

Table 1. Table of Species with 25 Largest Total Expenditures, 1993–2016

Species	Total Spending 1993–2016 (millions 2017\$)	Percentage of Total Expenditures
Chinook salmon ^a	4,543	18
Steelhead ^a	3,581	14
Coho salmon ^a	899	3
Sockeye salmon ^a	777	3
Bull trout	748	3
Pallid sturgeon	735	3
Steller sea lion ^a	730	3
Red-cockaded woodpecker	583	2
Desert tortoise ^b	482	2
Chum salmon ^a	381	1
Northern spotted owl	381	1
Wood stork	381	1
Bald eagle ^b	354	1
Southwestern willow flycatcher	323	1
Coastal California gnatcatcher	293	1
Piping plover ^b	278	1
Indiana bat	262	1
Razorback sucker	260	1
North Atlantic right whale ^a	239	<1
Louisiana black bear ^b	237	<1
West Indian manatee	233	<1
Rio Grande silvery minnow	232	<1
Grizzly bear	224	<1
Colorado pikeminnow	219	<1
White sturgeon	203	<1

^a Dropped from spending regressions due to National Marine Fisheries Service listing missing USFWS covariates such as range area.

^b Dropped from spending regressions due to inaccurate range information (i.e., reported range corresponds exactly to state boundaries).

1.2. Other USFWS Covariates of Listed Species

Since 1983, the USFWS has assigned priority numbers, intended to guide funding allocations, to each species. The priority number is a composite of three factors: degree of threat, potential for recovery, and taxonomic distinctiveness. These three factors are aggregated lexicographically into a single priority number that takes a value between 1 and 18, with the lexicographic ordering first prioritizing the degree of threat, followed by the potential for recovery, and last the taxonomic distinctiveness. In addition, USFWS assigns a “conflict” code, indicating whether or not the USFWS has determined that protection of the species conflicts with economic development. These priority numbers do

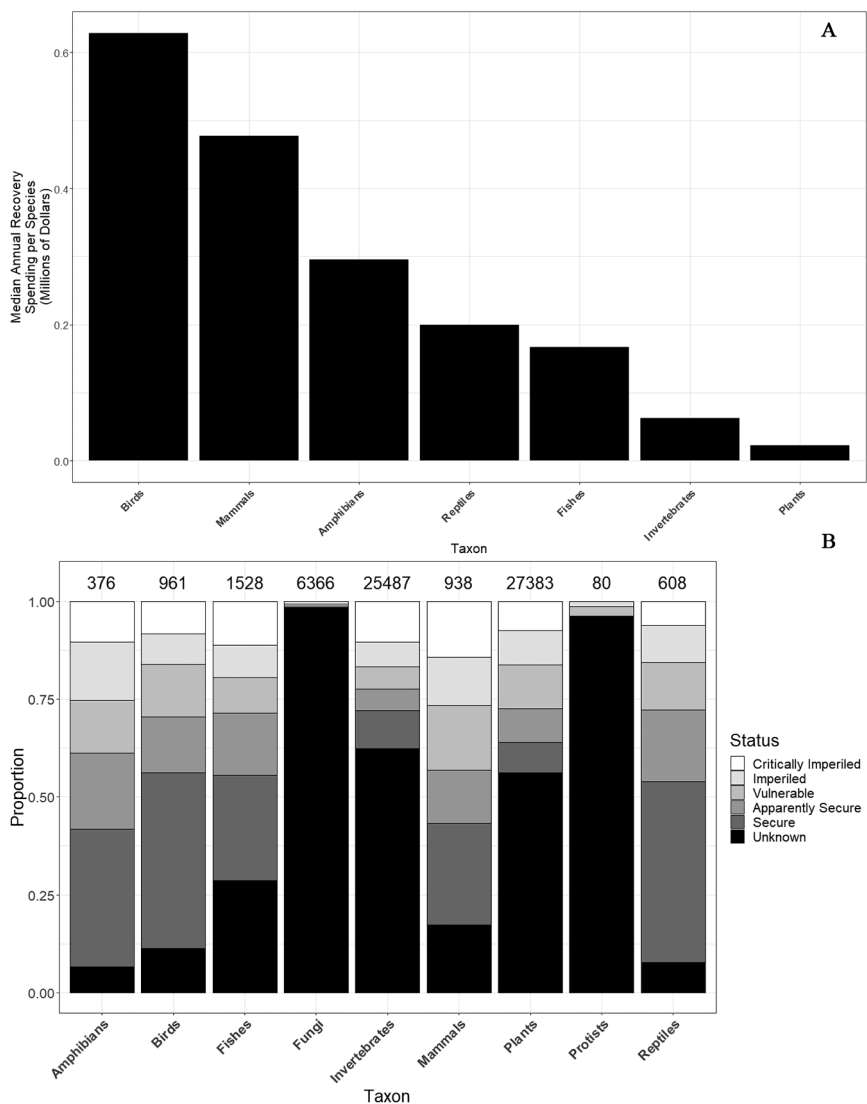


Figure 1. *A*, Annual spending by taxonomic grouping for the median species within each group. (Mean values are substantially higher for birds and fish because of the highly concentrated spending on a few species, as shown in table 1.) *B*, Distribution of assessed conservation status for 64,589 North American species and subspecies in the NatureServe data set. Numbers above bars give the total number of species within each group.

change over time, but such changes are uncommon; 72% of species have a single priority number throughout the 21-year record. Historical priority numbers were obtained from species recovery reports, which are reported biennially and extrapolated to intervening years.

Additional information on listing and delisting dates, current federal listing status, lead USFWS administrative regions, species occurrence in USFWS administrative regions, and geographic range were retrieved from the USFWS's Environmental Conservation Online System. Species ranges offered by the USFWS represent the geographic area where a species is known or suspected to occur and omit historical areas of occurrence. Ranges that were not defined by the USFWS at a resolution finer than a US state level were excluded from analysis as possibly not providing an accurate assessment of the true species range. Data for distinct population segments of the same species were aggregated by taking the minimum priority number rank, the earliest listing date, and the mode for listing classifications (i.e., endangered or threatened) and lead regions across populations of the same species.

1.3. Endangerment of Listed and Nonlisted Species

The conservation status and taxonomic information for listed and nonlisted species was obtained from the US nonprofit NatureServe. NatureServe tracks 53,417 full species and 11,172 subspecies and varieties found within North America. Conservation status ranks are assigned on a 5-point scale of imperilment ranging from "secure" to "critically imperiled" for extant species, with additional categories of "possibly extinct," "presumed extinct," and "unknown." When a species included a range of imperilment categories, reflecting uncertainty in a species' conservation status, we took the most conservative (i.e., most imperiled) conservation rank. Although NatureServe periodically revises its assessment of species, we have access only to the most recent assessment for each species. The most recent assessments were conducted between 1985 and 2019, with the vast majority occurring between 1996 and 2005 (mean date is 2003).

The details of the NatureServe ranking strategy are complex, but broadly speaking it is a weighted average of three components. The most important is rarity, defined by both population size and range extent. The other two components are the anthropogenic threat to populations and the short-term (10–50-year) and long-term (~200-year) trends in population or extent (Faber-Langendoen et al. 2012). A breakdown of conservation status by different taxonomic categories is shown in figure 1B. An important aspect of the data set revealed clearly in figure 1B is the large number of species with unknown status. Sixty percent of species in the data set have an unknown status. These are almost entirely invertebrates (41%), plants (40%), and fungi (16%). This unknown status is not simply missing information, but informative in the sense that the lack of scientific attention itself reveals this set of species to be not highly prioritized in the conservation process.

1.4. Google N-grams for Listed and Nonlisted Species

In their original studies, Metrick and Weitzman emphasized both the importance and the difficulty of measuring the utility value of species to humans, which they proxied

using taxonomic groupings (i.e., mammal, bird, amphibian, reptile, and fish) and average body length (1996, 1998). In this analysis, we preserve the taxonomic groupings but add a proxy for the utility value of species using the frequency that species' names appear in books published in the English language. The benefits of this measure are that it integrates over many different ways in which the existence of a particular species may provide utility—a species may be culturally significant, may be of scientific interest either because of unique biological characteristics or the role it plays within an ecosystem, or may simply be cute, interesting, beautiful, or majestic. Some of these characteristics map only imperfectly onto body length but would all likely influence the frequency that the species is written about. An additional benefit of these data is that they vary over time, allowing us to estimate the within-species effect of changing prominence on spending.

The Google Books Ngram Viewer provides the frequency of particular words or phrases within a corpus of over 28 million books in eight languages (Google 2020). Google Book's English 2019 corpus consists of over 16 million books and 1.9 trillion words published in the English language between 1470 and 2019 (Michel et al. 2011). To obtain n-gram frequencies for species names in Google Book's English 2019 corpus, case insensitive searches were performed for all scientific and common names in the NatureServe and ESA data sets for the years between 1800 and 2016. When a name failed to return valid data between the years 1800 and 2016 (Google Books Ngram Viewer only offers frequencies for words and phrases that occur in at least 40 books), the name was assigned a frequency of zero across all years. Some limitations of the n-gram measure are that it captures only English-language media and includes only material published in books, excluding some popular media such as magazines, websites, or newspapers.

We treat scientific and common name n-grams separately because common names pose particular challenges. Species may be known vernacularly by multiple names or may lack a common name altogether. To help account for generic common names (e.g., "blackberry" for *Rubus ostryifolius*; "a millipede" for *Trigenotyla blacki*) and common names with additional meanings or uses (e.g., "small blue" for *Philotiella speciose*; "British soldier" for *Cladonia cristatella*), we discarded common name n-grams with a standardized frequency 10 times greater than the standardized scientific name n-gram frequency for a given species.

N-gram frequencies were aggregated to the species level (for the listing decision) and the species-by-year level (for the spending decision). Species-level n-gram frequencies are the average from 1950 to 2016, except for listed species. For listed species, we take the average frequency from 1950 to 10 years before the listing decision, to avoid any chance that publications generated by the listing decision itself contribute to the n-gram value. The species-by-year n-gram frequency used in the spending regression is the lagged five-year rolling mean, which smooths out idiosyncratic year-to-year variation and allows for a delayed effect on spending patterns. Because raw n-gram frequencies are extremely small, all values are standardized before analysis.

1.5. Evolutionary Distinctiveness

We used two variables to represent species uniqueness. The first one, Evolutionary Distinctiveness (ED) describes species' relative contribution to the total evolutionary history, or phylogenetic diversity (PD). For a group of species comprising the extant descendants of a common ancestor (clade), PD of the clade is the sum of all branch lengths of the phylogenetic tree, measured in millions of years. ED is the "fair proportion" (Hartmann 2013) of the total PD assigned to an individual species in that clade, with the length of each branch of the phylogeny divided equally among all species to which it is ancestral.

We used published ED scores for mammals (Gumbs et al. 2018), amphibians (Isaac et al. 2012; Safi et al. 2013; Gumbs et al. 2018), birds (Jetz et al. 2014; Gumbs et al. 2018), reptiles (Gumbs et al. 2018), and plants (Potter 2018), which were obtained from the EDGE (Evolutionary Distinctive and Globally Endangered) of Existence program (Zoological Society of London 2008). Overall, we collected ED scores for 244 out of 309 amphibians (79%), 649 out of 768 birds (85%), 374 out of 439 mammals (85%), 328 out of 342 reptiles (96%), and 319 out of 19,092 plants (1.7%) in our list. No ED scores were available yet for fish, fungi, invertebrates, or protists in our study. Because the limited data availability for plants could imply a strong selection effect, we omit them from the regression that includes the ED score, considering only mammals, birds, amphibians, and reptiles.

The second metric to describe species uniqueness is the number of species within a genus, a variable originally used in Weitzman's studies (1992, 1993). Compared to ED, data for this metric cover significantly more species in our list. Decisions about assigning species to Linnean ranks above the species level are subjective, and genera size can vary simply based on the tendency of particular taxonomists toward either lumping or splitting (Darwin 1857; Laurin 2010). However, because genera have historically been defined based on observable characteristics of species, it may be that this measure actually captures aspects of distinctiveness that are more salient to humans. The data on the number of species in a genus were obtained from two databases: the Integrated Taxonomic Information Service (ITIS) and NCBI (National Center for Biotechnology Information) taxonomy database, using *taxize* package in R.

The noisiness of genera size as a proxy for genetic distance is shown in supplementary figure 1 (supplementary figs. 1, 2 are available online). Although the relationship is in the expected direction (species in smaller genera tend to be more evolutionarily distinct), it explains only a very small fraction of the variance ($R^2 = 3.6\%$), suggesting that genus size is only a very imperfect measure of the genetic distinctiveness of a particular species.

2. MODEL AND RESULTS

2.1. Listing Status

We model a species' listing status at the time of its NatureServe assessment as a binary outcome using a logistic regression. Although in situations with random treatment assignment and fixed effects the linear probability model is sometimes preferred (Lancaster

2000), these conditions do not apply here. In our case the unconditional probability of listing is low, about 2.5%. In addition, because a goal of the analysis is estimating the effects of climate change, which requires predicting out of sample, we use a model that returns strictly positive predicted probabilities.

The binary listing variable is modeled as a function of taxon (specifically amphibian, bird, fish, fungi, invertebrate, mammal, plant, protist, or reptile), the NatureServe conservation assessment status, standardized n-grams for both the scientific and common name of the species, and two different measures of taxonomic uniqueness. To ensure that the assessed conservation status corresponds to the listing status at the time of assessment, we code species as “unlisted” if they have never been listed, were delisted before the year of the conservation assessment, or were listed more than 10 years after the assessment.

To capture the evolutionary uniqueness of a species, we first include the number of species within the genus. This is a very rough proxy for genetic distinctiveness (supplementary fig. 1) but is available for almost all species. In a second regression we also report the more precise measure of evolutionary distinctiveness, which directly measures phylogenetic uniqueness of species but is only available for a subset of species. Because these genetic data are only available for birds, mammals, amphibians, and reptiles, the second regression (“Few Taxa”) is limited to just those species.

Standard errors in all regressions are estimated using 250 clustered bootstraps, clustering at the family level (the taxonomic level above genus), which allows for correlation of the residuals between related species. There are 1,933 families in the data set, and the median family has five species.

Results are shown in the first column of table 2. Like Metrick and Weitzman (1996, 1998), we find strong evidence that the probability of listing changes with conservation status. The probability of a species appearing on the ESA list at the time of assessment decreases monotonically from the most (status = 1, critically imperiled) to least (status = 5, secure) endangered. Coefficients in table 2 give the difference in listing probability compared to critically imperiled status. Table S1 (tables S1, S2 are available online) gives the difference in regression coefficients between different conservation status levels and shows that these differences are almost all significantly different from each other at the 5% level.

However, also like Metrick and Weitzman, we find evidence that factors associated with species’ utility value to humans affect listing. Although differences in listing probability do not differ significantly between vertebrate species (i.e., mammals, birds, fish, amphibians, and reptiles), plants and invertebrates are much less likely to be listed. However, we do not see an effect of either scientific or common name n-grams on listing probability.

The two variables related to the distinctiveness of a species show an interesting pattern. In the full model including all taxa, species in smaller genera are more likely to be listed. Increasing the number of species in the genus by 10% decreases the probability of listing by 0.42 percentage points at the population mean, roughly a 16% decrease relative

Table 2. Results of Listing Decision Regression

	Dependent Variable: Listed	
	All Taxa, No ED (1)	Few Taxa, Including ED (2)
Taxa (dropped = Mammals):		
Amphibians	-.607 (.326)	-1.137 (.509)
Birds	.419 (.393)	-.450 (.373)
Fish	.057 (.330)	
Fungi	-15.615*** (.552)	
Invertebrates	-2.441*** (.475)	
Plants	-.951*** (.311)	
Protists	-15.524*** (.707)	
Reptiles	.066 (.443)	-.466 (.494)
Conservation status (dropped = Critically imperiled):		
Imperiled	-1.363*** (.125)	-.335 (.318)
Vulnerable	-2.985*** (.167)	-1.923*** (.483)
Apparently secure	-4.557*** (.361)	-3.093** (1.227)
Secure	-7.201 (5.179)	-5.613 (6.159)
Extinct	-2.106*** (.545)	-.206
Probably extinct	-1.264*** (.267)	.435 (4.127)
Unknown	-6.003*** (.344)	-3.189 (4.378)
Other covariates:		
Common n-gram	.642 (.436)	.447 (.579)
Scientific n-gram	.021 (.219)	.059 (.192)
Genus size (logged)	-.174*** (.046)	-.262 (.161)
ED (logged)		.020 (.275)
Observations	53,688	1,501
Log likelihood	-3,158.321	-241.391
Akaike information criterion	6,354.641	512.782

Note. Dependent variable is a binary variable indicating whether the species is listed under the ESA or not at the time of conservation assessment, and is modeled using a logistic regression. Status gives the conservation status of the species as assessed by NatureServe, with lower numbers corresponding to higher endangerment. Standard errors are block bootstrapped, clustering at the family level. ED = evolutionary distinctiveness.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

to the average listing probability of 2.5%. This effect is largely driven by plants and invertebrates, which have both larger and more variable genera size. Column 2, however, shows that there is no evidence that evolutionary distinctiveness—the unique evolutionary history content contained in the “library” of a particular species (Metrick and Weitzman 1998)—plays a role in the listing decision. Because these genetic data are only available for birds, mammals, amphibians, and reptiles, the regression in the second column of table 2 (“Few Taxa”) is limited to just those taxa groups.

Figure 2 converts the coefficients for the full model in table 1 into a probability of listing for each taxonomic group and NatureServe conservation status rank. It shows the strong preference for vertebrate species over plants or invertebrates apparent in the listing

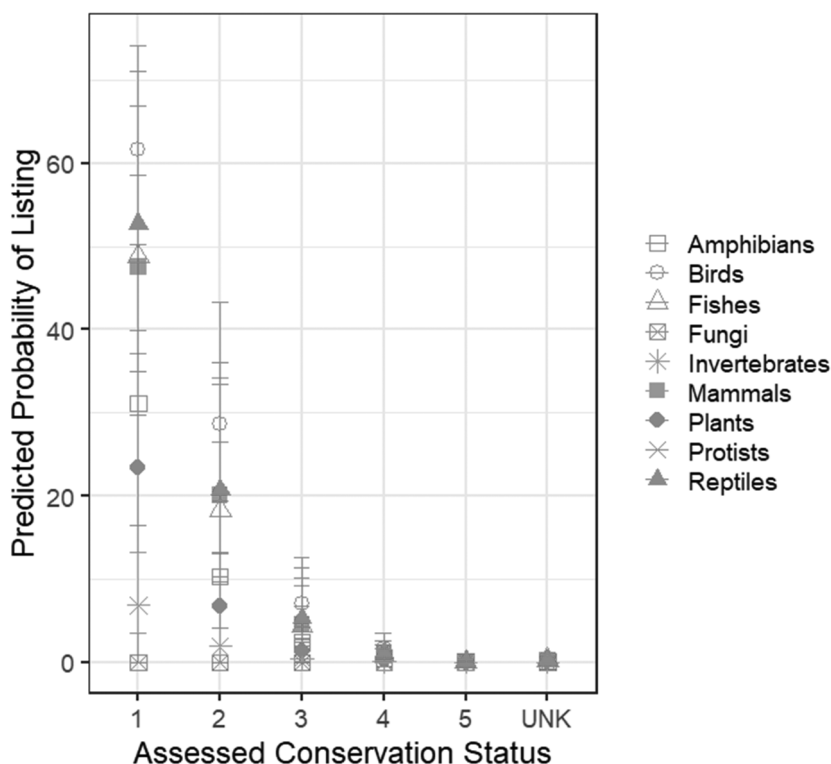


Figure 2. Probability of a species appearing on the ESA list at the time of conservation assessment, implied by coefficients in table 2 (col. 1) for each taxonomic group and NatureServe conservation rank. Error bars give the 95% confidence interval. Probabilities are estimated at the median value of model covariates (n-gram value and genera size) for each taxon by status combination. UNK = unknown.

decision, as well as the steep decline in listing probability with improvements in the assessed conservation status.

2.2. The Spending Decision

The logarithm of spending by USFWS on a particular species in a particular year is regressed on a set of variables that capture the taxonomic group, the listing status (specifically endangered, threatened, or extinct/probably extinct), the spatial extent of the species (both its range and the number of USFWS regions in which it occurs), the notability of the species (measured by the *n*-gram five-year rolling mean), the number of species in the genus, number of years since listing, and a set of variables capturing USFWS's own prioritization. The USFWS covariates include the three components of the priority number (degree of threat, species taxonomic rarity, and recovery potential) and an indicator variable capturing USFWS's assessment of whether conservation of the species conflicts with economic development. Standard errors are clustered at the family level in all regressions.

The inclusion of fixed effects at the species level helps control for unobserved factors that may affect spending but limits the set of variables that can be identified. We navigate this trade-off by presenting two models. A correlated random effects model ("full model") estimates the effect of all variables, including time-invariant species-level characteristics. Species-level unobservables are modeled through a random species effect, nested within families to allow correlation between related species. The fixed-effect model ("SpeciesFE") removes all time-invariant variation between species. This potentially improves confidence in the causal effects estimated but limits the effects that can be estimated only to those that change over time within a species, including USFWS prioritization, ESA listing status, and *n*-grams. Both models include year fixed effects that control for common variation over time.

Table 3 gives results of the spending regressions. We find evidence for a range of factors influencing patterns of spending. First, vertebrates receive, on average, more funding than plants or invertebrates, though there is no significant difference among vertebrate groups. Second, there is no indication that listing status (i.e., threatened vs. endangered) influences spending. It is possible that spending might influence conservation status, although several prior studies have looked at the relationship between cumulative lagged spending and species recovery, as measured in the biennial species recovery reports, with mixed results (Miller et al. 2002; Kerkvliet and Langpap 2007; Gibbs and Currie 2012).

Like previous papers (Metrick and Weitzman 1996; Langpap and Kerkvliet 2010), we find that the "conflict" indicator variable is both large and statistically significant in the full model, meaning that species where conservation has been identified as being in conflict with economic development tend to receive significantly more spending. Unlike much previous work, we also find effects of other elements of USFWS prioritization on spending (cf. Simon et al. 1995). The USFWS ranks the importance of the priority number criteria in the order threat, then recovery potential, then rarity, and we see some evidence of that in our estimates. The effect of a one-unit change in threat level (on a 3-point scale) is around a 15% increase in spending (statistically significant in both models).

Table 3. Results of Spending Decision Regression

	Dependent Variable: Annual Spending (logged)	
	Correlated Random Effects Model Full Model	Ordinary Least Squares Species FEs
Taxon (dropped = amphibians):		
Birds	.858 (.522)	
Fish	-.274 (.531)	
Invertebrates	-1.685*** (.501)	
Mammals	.254 (.534)	
Plants	-1.841*** (.491)	
Reptiles	-.413 (.598)	
Status (dropped = endangered):		
Threatened	.044 (.083)	.108 (.180)
Extinct	-1.008*** (.206)	
Priority number:		
Threat	.161*** (.038)	.145** (.068)
Rarity	.047 (.061)	.034 (.163)
Potential	.092** (.043)	.115 (.071)
Conflict	.265*** (.048)	.155 (.105)
Geographic factors:		
FWS regions	.792*** (.078)	
Range area (logged)	.148*** (.017)	
Other covariates:		
Scientific n-gram	.087** (.042)	-.028 (.064)
Common n-gram	.067** (.033)	.123** (.048)
Years listed (logged)	-.016 (.024)	-.029 (.075)
Genus size (logged)	-.043 (.027)	
Random effects	Species nested in family	NA
Fixed effects	Year	Species, year
Observations	20,011	22,995
R ² (full model)		.724
R ² (projected model)		.002
Log likelihood	-33,734.490	
Akaike information criterion	67,558.980	

Note. Dependent variable is logged species by year spending. Spending data span 1993–2016. The R^2 of the projected model gives the fraction of variance explained by the regressors after removing the fixed effects. Standard errors are clustered at the family level.

* $p < .1$.

** $p < .05$.

*** $p < .010$.

For recovery potential that falls to around 10% (significant in the full model but not the species fixed-effect model) and to a statistically insignificant 3% for species rarity.

Several variables new to this analysis have both large and statistically significant effects on spending. Widely distributed species, measured both by range area and the number of USFWS regions in which the species occurs are associated with higher spending. We also find an association between species' prominence in English-language books and spending. These effects are empirically small though—typical within-species variation in the scientific n-gram would lead to changes in spending on the order of 0.5%, or about \$200 for the median species. We do not find any systematic effect of years since listing or phylogenetic distinctiveness (i.e., genus size).

Using the full models, we calculate an expected present cost of 100 years of managing listed species under the ESA. The cost (C) of managing a listed species in taxonomic group g is calculated via simulations from the full random effects model as follows:

$$C_g = \sum_{t=0}^{100} \frac{1}{N_g} \sum_{n \in g} \left(e^{(\hat{\beta}_0 + \hat{\beta}_g + \hat{\beta} \bar{X}_g + \hat{\beta}_t \log(t) + \gamma_n + \varepsilon)} \right) \cdot DF_t, \quad (1)$$

where the β terms are draws from the multivariate distribution of model coefficients, $\bar{X}_g$ is the group-specific average of model covariates, γ_n is the estimated random effect for species n in group g , N_g is the total number of species in taxonomic group g , and ε is a draw from the residual distribution. The mean and distribution of C_g is calculated based on 1,000 samples of the estimated parameter and residual distribution.

The term DF_t is the discount factor in year t , calculated using the declining discount rate schedule given in Weitzman (2001). There are a number of reasons why a declining discount rate may be appropriate for long-term projects such as species preservation (Arrow et al. 2013, 2014). In "Gamma Discounting," Weitzman points out that under uncertainty, the relative importance of the lower region of the discount rate distribution grows over time, an effect that can be approximated by a declining discount rate. The schedule he provides, calibrated to a survey of 2,160 economists, starts at 4% per year and declines steadily to 1% after 75 years. To account for the possibility of future delisting, we estimate the average annual probability of delisting from our sample at 0.09% (resulting from 24 recoveries and seven extinctions) and add this to the annual discount rate.

Results are given in table 4. It is important to note that these costs represent only direct public spending on listed species and assume stability of policy priorities and public preferences into the future. There are also the opportunity costs of restrictions on land use and other activities that accompany listings that are not bounded here but could be large. These are discussed further in section 3.3.

What, in sum, do our findings reveal about the listing and spending decisions under the ESA compared to Metrick and Weitzman (1996, 1998)? Most significantly, while the previous papers showed some evidence for preferences in both listing and spending for some vertebrate groups over others, we do not find evidence for prioritization among

Table 4. Present Costs of Direct Spending on Listed Species for 100 Years Following Listing (million \$) with 95% Confidence Interval

	Central Estimate	Lower Bound (2.5%)	Upper Bound (97.5%)
Amphibians	39.5	32.2	50.2
Birds	107.6	99.3	117.6
Fish	57.9	52.6	64.7
Invertebrates	8.4	7.9	9.1
Mammals	74.1	68.3	80.6
Plants	3.5	3.3	3.6
Reptiles	61.1	52.0	72.8

vertebrate groups but do find large preferences for vertebrate species over the much larger groups of plants and invertebrates that were not considered in previous papers. We also find associations between USFWS prioritization and spending, particularly the large effect of the “economic conflict” variable, but also an important role for the “degree of threat” indicator, the most important factor in the lexicographic prioritization scheme used by USFWS. We also document an important role of species distribution, both absolute range area and the number of FWS regions it covers in determining spending. Finally, the null findings are also notable. We do not find an effect of species n-grams on listing probability and only a very small effect on spending. We also do not see evidence that species representing particularly distinctive evolutionary histories—the most unique of the genetic libraries described by Weitzman in the Noah’s Ark problem (1998)—are prioritized for ESA listing.

3. CHANGE IN DIRECT ESA EXPENDITURES IMPLIED BY CLIMATE CHANGE

Climate change will affect species everywhere and, for some subset of species, will pose an existential threat. Rising temperatures could therefore imply a growing fiscal and regulatory burden for species protections in the United States. The magnitude of this burden has not previously been constrained, but the results shown in tables 2 and 3 can be used for a rough calculation of climate change costs, assuming that priorities for species protection persist into the future. This assumption is large but can be at least partly justified by the observation that analyses of ESA spending data conducted almost 25 years apart reveal similar patterns, such as higher spending on species in conflict with economic development (Metrick and Weitzman 1996, 1998).

3.1. Effects of Climate Change on Species Extinction

There is now a large literature in ecology estimating the number of species at risk of extinction at different levels of warming. The vast majority of this literature uses the climatic range of the current distribution of species and examines how this area will expand or shrink with warming. Some studies incorporate limits on the rate of species movement,

but they typically exclude factors such as species interactions or landscape dispersal barriers, or ecologically important but species-specific factors such as breeding grounds, all of which may modulate the impact of climate change (Urban 2015). Species that lose more than a threshold area of their range are projected to be at risk of extinction. A 2015 meta-analysis of 131 published studies showed that species at risk of extinction increase from approximately 3% at 1°C of warming to almost 20% at 5°C (Urban 2015).

For this paper, we reestimated the temperature-extinction relationship using the subset of 191 estimates from 32 separate studies focused on North America in the Urban (2015) data set. Each estimate is one result from an ecological study giving the fraction of species estimated to be at risk of extinction at a particular level of warming. A single paper may report multiple estimates that differ in terms of the level of warming, the set of species considered (e.g., birds vs. mammals vs. invertebrates), the dispersal model used (assumptions about how species might move as the climate shifts), and the threshold range contraction defining when a species is defined as at risk of extinction. Within the 191 estimates, 98 (51%) assume universal dispersion (i.e., no geographic or speed constraints on movement to a new area), 25 (13%) assume some species-specific rate of movement limiting dispersion, 21 (11%) assume that species can only move into areas contiguous to their current range, and the remaining 47 (25%) assume no species movement.

Using the North American studies, we estimate the regression:

$$\text{Ext} = \beta_1 \Delta T + \beta_2 \Delta T \cdot \text{Threshold} + \varepsilon,$$

where Ext is the proportion of species estimated to be at risk of extinction and ΔT is the rise in temperature since the pre-industrial period. Threshold is the proportion of range area lost before the species is “counted” as functionally extinct—many studies use a 100% threshold (i.e., the climatic range ceases to exist), but others vary between 80% and 100%. We allow this choice to modify the temperature-extinction relationship. Standard errors are clustered at the author level, allowing for correlation between estimates from the same study, as well as between different studies by the same author. The regression is weighted by the number of species each study examined to estimate the fraction of species at risk of extinction.

Results are given in table 5 and show the expected increase in extinction risk with warming, as well as the effect of the threshold variable modifying this relationship in the expected direction. We test, but do not find evidence for, both a quadratic effect of temperature change ($p = .71$) and heterogenous effects of warming on plants, vertebrates,

Table 5. Results of Regression of Fraction of Species at Risk of Extinction on Warming Level

	Parameter	<i>p</i> -Value
ΔT	.380	.004
$\Delta T \times \text{Threshold}$	-.377	.005

and invertebrates (a Wald test of the joint significance of taxa-temperature interaction terms fails to reject the null hypothesis, $p = .29$).

Supplementary figure 2 displays the estimated effect and uncertainty range for a threshold value of 90%. We conservatively chose a 90% value to represent the fact that species are of conservation concern long before they are on the brink of extinction; the International Union for the Conservation of Nature (IUCN) criteria for critically endangered, for example, includes a reduction in population greater than or equal to 80%. We find that the fraction of species at risk of extinction in North America increases from 4% at 1°C of warming to 20% at 5°C, a close quantitative match to the global average values reported by Urban (2015).

3.2. Effects of Climate Change on Listing and Spending

Combining the expected effect of warming on extinction risk, the probability of listing conditional on conservation status, and the average level of direct spending conditional on listing, it is possible to generate a rough, ballpark estimate of the fiscal burden in terms of direct spending on the ESA implied by climate change, assuming that future public preferences over endangered species protection resemble those of the last 25 years. There are two main challenges in making this estimate. First, the mapping from “risk of extinction” as defined in ecological studies of climate impacts, to the NatureServe conservation status is unclear. We proceed under the working assumption that the species threatened with extinction by climate change will have a conservation assessment of 1 (“critically imperiled”). This means that, by construction climate change has no effect on the probability of listing for species already critically imperiled. Range contractions caused by climate change for these already imperiled species might substantially increase the probability of listing. It is also possible though, in theory, that climate change could lead to range expansions for some of these species, lowering the probability of listing. In either case, we are not able to resolve this effect given the coarse nature of conservation status information for nonlisted species.

A second ambiguity concerns the future status of the large number of species with unknown status (61% of the NatureServe data set). The vast majority (98%) of these species are plants, invertebrates, or fungi. Given the demonstrated lack of attention and scientific knowledge about these species, it is unclear whether any existential threat posed by climate change would be sufficient to propel them onto the Endangered Species List. Accordingly, our preferred results err conservatively by including only the set of species with a current conservation assessment as potentially listable, though we also present results that include the full set of species for comparison. Details on the simulations used to estimate these effects is given in the appendix (available online).

Figure 3A shows the expected increase in the number of critically imperiled species due to climate change, as well as the subset of those that would be expected to be listed, excluding any species where status is currently unknown. The linear increase with warming follows from the estimated linear effect of warming on extinction risk (supplementary fig. 2). The strong selection at the listing stage is also apparent—on average less than 20%

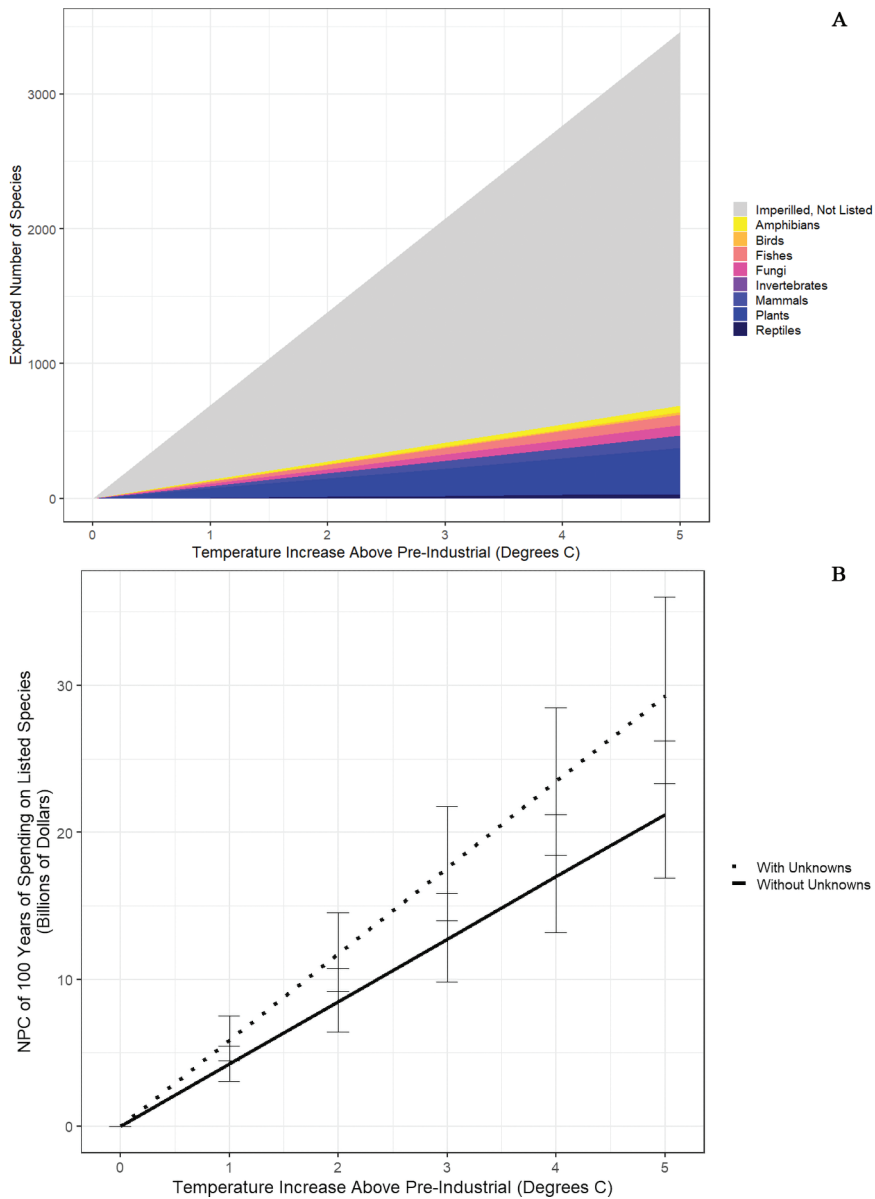


Figure 3. *A*, Expected increase in number of species that are critically imperilled due to climate change with warming, as well as the expected subset that would be listed (in colors). Includes only species with currently known conservation status as potentially listable. *B*, Net present cost of committed direct spending implied by climate-change-induced listings, both including and excluding the set of species with unknown conservation status. Error bars show the 95% confidence interval based on 1,000 simulations.

of the species critically imperiled by climate change would be listed. The relative representation of different taxonomic categories among listed species depends both on their absolute prevalence (there are far more invertebrates and plants than mammals or birds) as well as the differential probability of listing (invertebrates are far less likely to be listed for instance, fig. 2). A warming of 2°C would increase the number of species listed by 280 in expectation, increasing to 690 at 5°C of warming. An additional 1,100 (2°C) to 2,800 (5°C) species, mostly plants and invertebrates, will be critically imperiled but remain unlisted. Including the additional 38,430 species with unknown conservation status increases the expected number of additional listings to 610 and 1,540 under 2°C and 5°C, respectively. Depending on greenhouse gas emissions over the twenty-first century, 2°C might be reached any time between 2050 and 2100 and 5°C might be reached by 2100 under a high emissions trajectory (Collins et al. 2013).

Figure 3B gives the total expected increase in protection costs with warming, both including and excluding species with unknown conservation status. Error bars give the 95% confidence interval accounting for four sources of uncertainty: the effect of warming on extinction risk, which species are affected by climate change, uncertainty in the probability of listing, and uncertainty in spending, conditional on listing. Our preferred estimate that excludes species with currently unknown conservation status implies a steady increase in committed direct ESA spending with warming, reaching \$4.3 billion at 2°C and \$21.2 billion at 5°C. Although including unknown species into the analysis more than doubles the expected number of listed species, spending is only ~40% greater because the majority of these species are plants and invertebrates that receive relatively few resources (table 3).

How large are these values? Certainly not large compared either to the total costs of climate change or the discretionary spending of the US federal government. But relative to historic spending on endangered species, these represent a substantial additional burden. For the 1,613 listed species that appear in our data set, the present cost of 100 years of spending totals just under \$40 billion (based on the taxa averages given in table 2). Unmitigated climate change therefore represents an increase of somewhere between 50% and 75% of all the resources committed to direct spending on endangered species protection since the beginning of the ESA in the late 1960s. This ignores any effect of climate change on the listing of species that are already highly imperiled, any interactions of climate change with other threats to species such as habitat destruction or invasive species, or the possibility that climate change might increase average spending on listed species. For these reasons, the estimate could be considered a lower bound.

3.3. Other Listing Costs

Direct conservation expenditures are only one part of the social cost of species conservation. The ESA protects listed species in two main ways. First, the agency designates critical habitat, which requires federal agencies to consult with USFWS before making decisions that could threaten listed species (ESA, sec. 7; Langpap et al. 2018). In some states, critical habitat designation also triggers additional state and local review of land use

permitting (Auffhammer et al. 2020). Not all species have critical habitat designated—as of 2020 it applied to about 850 out of over 1,600 listed species. Second, the “no-take” provision of the ESA (sec. 9) prohibits any action that harms endangered species, interpreted to include indirect harm through habitat modification (Langpap et al. 2018). This section, which applies to all listed species, can directly limit private land development and otherwise restrict economic activity.

The total cost of these provisions as a whole is not well established (Plantinga et al. 2014). It is clear that for some species, no-take and critical habitat provisions impose large costs on particular landowners or sectors (Sunding and Terhorst 2014; Auffhammer et al. 2020; Frank 2020). Melstrom (2021) estimates negative effects on the order of 4% from critical habitat listing on the profits and value of dryland agriculture, with no effects on irrigated farmland. Other have shown that the aggregate welfare consequences of these restrictions can depend sensitively on the specifics of land, labor, and product markets, as well as the interaction with other local or state restrictions on land use (Murray and Wear 1998; Quigley and Swoboda 2007; Sunding and Terhorst 2014). The ESA contains very few provisions that allow for cost-effective management of species protection (Plantinga et al. 2014), and there is evidence both for highly heterogeneous protection costs among landowners (Sunding and Terhorst 2014) and steeply increasing marginal costs at high levels of protection (Langpap et al. 2018). Taken together, these suggest that the costs from no-take and critical habitat constraints, compared to more cost-effective management, may be substantial, at least in certain contexts.

The threat that climate change poses to biodiversity therefore has implications not just for direct spending on conservation but also for the opportunity costs that protection of endangered species requires. These costs are at present unquantifiable but should be understood as increasing the direct costs shown in figure 3B by an unknown but potentially large multiplier. The total adaptation costs for species protection in this second- or even third-best regulatory setting therefore come from (1) the increase in direct conservation spending estimated in the previous section, (2) the welfare costs of raising these funds from distortionary taxes, and (3) welfare loss from constraints on economic activity required for the protection of species that would not have been endangered in the absence of climate change. The last category may well be the largest, but given the limited evidence on the costs of ESA restrictions, it cannot currently be estimated. Instead it joins the long list of “known unknowns” in the enumeration of climate change costs (Pindyck 2013).

4. THE BENEFITS OF SPECIES PROTECTION AND RESIDUAL CLIMATE CHANGE DAMAGES

Interpreting the increase in listing and associated social costs of species protection with climate change as adaptive requires that the benefits of listing protections for these species exceed the social costs. A precise determination of net benefits of ESA species listing is impossible for a number of reasons, not least that the total social cost of listing is unknown, as discussed in the previous section. Here instead we attempt to roughly bound

this question through a review of stated-preference estimates (i.e., contingent valuation and choice experiments) of household-level willingness to pay for species protection.

We identify 48 estimates from 28 studies for 36 different listed species (details for each estimate in table S2). These 36 species are not a random sample of listed species. Mammals, birds, and fish are vastly overrepresented compared to plants (1), invertebrates (3), and amphibians (0). Moreover, there is a focus on well-known species such as the gray wolf or sea turtles and on species with high-profile conflicts around ESA protections, specifically the red-cockaded woodpecker and northern spotted owl. The studies consider the value of a number of different outcomes related to species conservation, which may not be mutually exclusive, from habitat conservation (13/48 species), including critical habitat designation (5/48), to population changes, including avoided losses (17/48) as well as population gains (26/48). The studies quantify the value of these outcomes via contingent valuation (30/48) or choice experiments (18/48), using household WTP as the metric of choice.

Studies report WTP in terms of either a one-time payment (six estimates) or an annual payment (42 estimates, details in table S2). Given the preponderance of annual WTP estimates, we compare annual aggregate WTP values to average annual ESA expenditures for each species. For studies reporting a one-time payment, we convert these amounts into equivalent annual payments assuming a 20-year time horizon and a 3% discount rate. Although these one-time payment studies do not always specify a time horizon, those that do tend to reference periods of between 10 and 20 years. For studies with annual WTP estimates, this approach assumes preference stability for species over time. The relevant literature suggests reasonably stable preferences over the medium run, with less stability over longer time frames (Skourtos et al. 2010).

Total aggregated WTP values are sensitive to the population of households considered as holding existence values. In our preferred estimate, we assign the same WTP value across all households considered in the original study (e.g., the whole United States for nationwide samples, individual states for studies conducted only with local or state residents, “Full Population” in table 6). In a sensitivity analysis (“Restricted Population” in table 6), we also calculate WTP over a smaller population, limited to people living within the USFWS species range (for terrestrial species) or within coastal counties in coastal states adjacent to its range (for marine species). We compare this aggregate WTP to average annual ESA spending on each species over the 1993–2016 period in table 6.

The final two columns in table 6 show the ratio of direct ESA spending to aggregate WTP values for particular species. For the vast majority of species, WTP is between one and two orders of magnitude larger than direct spending levels. The major exception is the salmonids, which stand out for receiving a very large fraction of spending on endangered species (table 1). Salmon are somewhat unusual among endangered species in that they have high recreational and commercial use values as well as unique spiritual and cultural value to indigenous peoples of the Pacific Northwest, perhaps explaining the disproportionate level of spending they receive relative to the WTP measures in table 6.

Table 6. Comparison of Aggregate WTP for Conservation of 36 Listed Species from Stated-Preference Estimates with Average Annual Direct ESA Spending over the 1993–2016 Period

Species	References	Avg Annual ESA Expenditures (millions of 2017\$)		Avg Annual Agg. WTP (millions of 2017\$)		Avg Ratio WTP/ESA Expenditures	
		Avg Annual ESA Expenditures (millions of 2017\$)		Full Population	Restricted Population	Full Population	Restricted Population
Birds:							
Bald eagle	Boyle and Bishop (1987), Stevens et al. (1991)	17.7	6,073	2,268	343.2	128.2	
Northern spotted owl	Rubin et al. (1991), Hagan et al. (1992)	15.9	18,216	1,347	1,146.2	84.7	
Whooping crane	Bowker and Stoll (1988)	6.2	11,958	3,338	1,934.3	540.0	
Mexican spotted owl	Loomis and Ekstrand (1997), Giraud et al. (1999)	4.2	8,726	834	2,054.2	196.3	
Fishes:							
Chinook salmon	Olsen et al. (1991), Loomis (1996), Wallmo and Lew (2012)	189.3	4,634	141	24.5	.7	
Southern CA steelhead	Wallmo and Lew (2016)	188.5	5,628	518	29.9	2.7	
Coho salmon	Olsen et al. (1991), Bell et al. (2003), Wallmo and Lew (2016), Lewis et al. (2019)	44.9	466	236	10.4	5.3	
Sockeye salmon	Olsen et al. (1991)	32.4	581	373	18.0	11.5	

Table 6 (*Continued*)

Species	References	Avg Annual ESA Expenditures (millions of 2017\$)		Avg Annual Agg. WTP (millions of 2017\$)		Avg Ratio WTP/ESA Expenditures	
		Full Population	Restricted Population	Full Population	Restricted Population	Full Population	Restricted Population
Chum salmon	Loomis (1996)	21.2	5,119	945	241.6	44.6	
Atlantic salmon	Stevens et al. (1991)	10.2	377	13	37.1	1.3	
Silvery minnow	Berrens et al. (1996)	10.1	638	42	63.3	4.2	
Colorado squawfish	Cummings et al. (1994)	9.1	481	169	52.7	18.6	
Shortnose sturgeon	Korchen and Reiling (2000)	2.4	931	78	381.1	32.0	
Smalltooth sawfish	Wallmo and Lew (2012)	.9	5,799	225	6,476.6	250.9	
Invertebrates:							
Black abalone	Wallmo and Lew (2016)	1.4	5,588	736	3,922.5	516.6	
Elkhorn coral	Wallmo and Lew (2016)	1.1	5,441	31	4,891.8	28.3	
Riverside fairy shrimp	Stanley (2005)	.5	524	23	1,135.1	50.7	
Mammals:							
Steller sea lion	Giraud et al. (2002), Lew et al. (2010)	30.4	9,232	-130	303.7	-4.3	
North Atlantic right whale	Wallmo and Lew (2012)	9.9	7,591	209	763.0	21.0	
Manatee	Solomon et al. (2004)	9.7	670	103	68.9	10.6	
Gray wolf	Chambers and Whitehead (2003)	8.4	647	270	76.7	32.1	
Northern Rocky Mountain wolf	Duffield (1991)	8.4	6,986	581	829.3	68.9	
Peregrine falcon	Korchen and Reiling (2000)	6.8	436	2,587	64.5	382.5	
Hawaiian monk seal	Samples and Hollyer (1990), Wallmo and Lew (2012)	3.4	5,067	906	1,471.3	263.0	

Humpback whale	Wallmo and Lew (2016)	3.2	14,917	2,710	4,705.6	854.8
Southern resident killer whale	Wallmo and Lew (2016)	2.4	6,882	234	2,812.4	95.7
Cook Inlet Beluga whale	Lew (2019)	2.4	5	5	2.2	2.2
Blue whale	Hageman (1985)	1.0	4,056	820	4,039.3	817.0
North Pacific right whale	Wallmo and Lew (2012)	1.0	7,899	150	8,033.9	152.7
California sea otters	Hageman (1985)	.9	573	531	671.1	622.0
Bighorn sheep	King et al. (1988)	.4	524	2	1,188.6	3.9
Gray whale	Hageman (1985), Loomis and Larson (1994)	.3	441	107	1,299.6	313.9
Reptiles:						
Loggerhead sea turtle	Whitehead (1992), Wallmo and Lew (2012)	8.3	4,045	614	489.1	74.2
Leatherback sea turtle	Wallmo and Lew (2012)	6.1	7,283	814	1,201.1	134.3
Hawksbill sea turtle	Wallmo and Lew (2016)	4.0	9,994	1,426	2,469.2	352.3
Plants:						
Johnson's seagrass	Wallmo and Lew (2016)	1.3	\$6,597	\$6,051.6	5,028.4	4,612.4
Average for all birds		11.0	\$11,244	\$1,947	1,369.5	237.3
Average for all fishes		50.9	\$2,465	\$274	733.5	37.2
Average for all invertebrates		1.0	\$3,851	\$264	3,316.5	198.5
Average for all mammals		5.9	\$4,395	\$606	1,233.2	242.4
Average for all reptiles		6.1	\$7,108	\$951	1,386.5	186.9
Average for all plants		1.3	\$6,597	\$6,052	5,028.4	4,612.4

Note. Specifics of each estimate are given in table S2. Most studies report annual WTP. Six estimates are one-time payments, which are converted into an annual equivalent amount assuming a 3% discount rate and 20-year time horizon. Aggregate WTP values depend on the population the values are aggregated over and so two population definitions are reported: one based on the study sample area ("Full Population") and one based on a restricted population limited to the species range areas ("Restricted Population").

For the vast majority of species, aggregate WTP is at least an order of magnitude larger than direct spending levels, using the preferred population estimates ("Full Population" columns). Even a much more restricted definition of population gives ratios greater than 1 for ~90% of species and WTP to spending ratios on the order of 10 to 100 times for many species ("Restricted Population" columns).¹

Given the paucity of evidence on the total social costs of listing, as well as the difficulty of comparing the exact good being valued in WTP estimates with listing protections, this should not be interpreted as proving that the benefits of listing exceed the costs. But it does give some sense of how large the indirect costs of listing would have to be before exceeding WTP. The unmeasured opportunity costs of species protection would have to be between one and three orders of magnitude larger than the direct ESA spending in order to exceed WTP. Alternate assumptions around the distribution of property rights over extant species would imply the use of willingness to accept measures that are often higher than WTP (Hanemann 1991; Tuncel and Hammitt 2014).

Total climate change costs arise both from the costs of adaptation and the climate damages remaining after adaptation (Cropper and Oates 1992). Given the historical evidence, it is not clear that all or even a majority of species threatened with extinction by climate change would be protected by listing, and it is possible that some of these species may go extinct, implying that residual damages may be substantial. The welfare estimates in table 6 cannot be extrapolated to estimate these costs, as the set of species likely to be critically imperiled by climate change but remain unlisted are disproportionately plants and invertebrates, of which there are almost no WTP estimates. Other residual welfare losses that are even harder to bound come from declines in species abundance or the extinction of local populations without full species extinction as well as the possibility that listed species may be at higher risk of extinction due to climate change.

5. CONCLUSIONS

This paper has estimated how revealed public priorities for conservation spending will interact with a changing climate over the twenty-first century. We find that these priorities appear to be fairly stable, at least over the medium term, as we document somewhat similar patterns to those identified by Metrick and Weitzman (1996, 1998) 25 years ago, particularly the prioritization of species that are more imperiled and in conflict with economic development. By expanding the analysis to include the large groups of plants and invertebrates, we also document a strong preference, in terms of both listing and spending, in favor of vertebrate species.

1. Note that the negative values for the Steller sea lion in the restricted sample are taken from results in the sample of local residents who were faced with a situation in which the change in population being valued was related to the closure of local fishing grounds. This closure would be associated with negative income shocks for local residents, offering an explanation for the negative WTP values.

Based on these patterns, and the estimated effect of climate change on US biodiversity, we estimate that unmitigated climate change could increase committed direct spending under the ESA by 50%–75% by the end of the century. Further costs will arise from restrictions on land use associated with the expected 700 additional listings. Several thousand species, mostly plants and invertebrates, will be critically imperiled by climate change but remain unlisted. The lost existence value associated with these species is difficult to estimate: while our survey of the WTP literature suggests that these values are large, at least compared to direct ESA spending, the vast majority of these studies focus on vertebrate species, so the generalizability to plant and invertebrate species is unclear. For the limited subset of species included in valuation studies though, we find that WTP for conservation greatly exceeds direct ESA spending, meaning listings of these species are welfare improving unless the indirect costs of listing are orders of magnitude greater than the direct expenditures.

Weitzman published extensively on both the biodiversity and climate change problems. His work on climate change focuses on the role that very low probability but high consequence climate damages can play in driving cost-benefit results (Weitzman 2009, 2012a). In his writing on the issue, Weitzman emphasizes the importance of deep structural uncertainties in the costs of climate change and urges researchers not to sideline these concerns in pursuit of a false sense of precision or objectivity: “The economics of fat-tailed catastrophes raises difficult conceptual issues that cause the analysis to appear less scientifically conclusive and more contentiously subjective than what comes out of an empirical CBA of more usual thin-tailed situations. But if this is the way things are with fat tails, then this is the way things are, and it is an inconvenient truth to be lived with rather than a fact to be evaded just because it looks less scientifically objective in cost-benefit applications” (Weitzman 2009, 18). The welfare consequences of ecosystem disruption and decline caused by unmitigated climate change are highly uncertain and potentially large, with some evidence that WTP for species’ existence may be fat tailed (see Conte and Kelly [2021] for a review). Difficulty of measurement, compounding uncertainties in the climate, ecological, and economic estimates, and structural uncertainty in how these goods enter into the utility function (Stern and Persson 2008; Weitzman 2012b; Drupp and Hänsel 2020; Bastien-Olvera and Moore 2021) mean that these costs are currently very poorly constrained in climate damage estimates.

While the implications for direct-spending commitments on conservation can be estimated, this only highlights the large unknowns in the ecological costs of climate change, particularly the opportunity costs associated with species protection and the lost welfare from residual ecological damages. The ESA case also serves as a reminder that public climate change adaptation more generally will occur within an existing framework of laws and regulations, many of which are inefficient and distortionary, or which interact in complex ways with preexisting market distortions. Climate change will interact with and in some cases exacerbate these distortions, meaning a full accounting of climate change costs should include the additional deadweight loss in these second-best settings.

Weitzman's writing on biodiversity came at a time of growing public concern over the global loss of species from deforestation and land-cover change, resulting in the Convention on Biological Diversity, signed in 1992 at the Rio Earth Summit at the same time as the Framework Convention on Climate Change. Global progress on both issues in the almost 30 years since has been decidedly mixed, with the result that the two will become increasingly intertwined over the future as climate change interacts with existing pressures from habitat loss, pollution, and invasive species to drive ecosystem change. Understanding the welfare costs of these changes poses large empirical and theoretical challenges. But this understanding is also essential if environmental economics is to better inform the collective social response to these problems.

REFERENCES

- Adamowicz, W., J. Louviere, and M. Williams. 1994. Combining revealed and stated preference methods for valuing environmental amenities. *Journal of Environmental Economics and Management* 26:271–92.
- Alexander, Kristina. 2010. Warranted but precluded: What that means under the Endangered Species Act (ESA). Congressional Research Service, Washington, DC.
- Ando, Amy Whritenour. 2001. Economies of scope in endangered-species protection: Evidence from interest-group behavior. *Journal of Environmental Economics and Management* 41 (3): 312–32.
- . 2003. Do interest groups compete? An application to endangered species. *Public Choice* 114 (1–2): 137–59.
- Arrow, Kenneth J., Maureen Cropper, Christian Gollier, Ben Groom, Geoffrey M. Heal, Richard G. Newell, William D. Nordhaus, et al. 2013. Determining benefits and costs for future generations. *Science* 341 (6144): 349–50.
- . 2014. Should governments use a declining discount rate in project analysis? *Review of Environmental Economics and Policy* 8 (2): 145–63.
- Auffhammer, Maximilian, Maya Duru, Edward Rubin, and David L. Sunding. 2020. The economic impact of critical-habitat designation: Evidence from vacant-land transactions. *Land Economics* 96 (2): 188–206.
- Barrage, Lint. 2020. The fiscal costs of climate change. *AEA Papers and Proceedings* 110:107–12.
- Bastien-Olvera, Bernardo A., and Frances C. Moore. 2021. Use and non-use value of nature and the social cost of carbon. *Nature Sustainability* 4:101–8.
- Bell, Kathleen P., Daniel Huppert, and Rebecca L. Johnson. 2003. Willingness to pay for local Coho salmon enhancement in coastal communities. *Marine Resource Economics* 18 (1): 15–31.
- Bellard, Céline, Cleo Bertelsmeier, Paul Leadley, Wilfried Thuiller, and Franck Courchamp. 2012. Impacts of climate change on the future of biodiversity. *Ecology Letters* 15 (4): 365–77.
- Benton, M. J. 1994. Diversification and extinction in the history of life. *Science* 268 (5207): 52–58.
- Berrens, Robert P., Phillip Ganderton, and Carol L. Silva. 1996. Valuing the protection of minimum instream flows in New Mexico. *Journal of Agricultural and Resource Economics* 21 (2): 294–308.
- Blumm, Michael C., and Kya B. Marienfeld. 2014. Endangered Species Act listings and climate change: Avoiding the elephant in the room. *Animal Law* 20 (2): 277–309.
- Bowker, J. M., and John R. Stoll. 1988. Use of dichotomous choice nonmarket methods to value the whooping crane resource. *American Journal of Agricultural Economics* 70 (2): 372–81.
- Boyle, Kevin J., and Richard C. Bishop. 1987. Valuing wildlife in benefit-cost analyses: A case study involving endangered species. *Water Resources Research* 23 (5): 943–50.

- Ceballos, Gerardo, Paul R. Ehrlich, Anthony D. Barnosky, Andrés García, Robert M. Pringle, and Todd M. Palmer. 2015. Accelerated modern human-induced species losses: Entering the sixth mass extinction. *Science Advances* 1 (5): e1400253.
- Chambers, Catherine M., and John C. Whitehead. 2003. A contingent valuation estimate of the benefits of wolves in Minnesota. *Environmental and Resource Economics* 26 (2): 249–67.
- Collins, Matthew, Reto Knutti, Julie M. Arblaster, Jean-Louis Dufresne, Thierry Fichefet, Pierre Friedlingstein, Xuejie Gao, et al. 2013. Long-term climate change: Projections, commitments and irreversibility. In *Climate change 2013: The physical science basis*. Contribution of Working Group 1 to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change, ed. Thomas F. Stocker, D. Qin, G. K. Plattner, M. Tignor, S. K. Allen, J. Boschung, A. Nauels, Y. Xia, V. Bex, and Guy F. Midgley, 1029–1136. Cambridge: Cambridge University Press.
- Conte, Marc N., and David L. Kelly. 2021. Understanding the improbable: A survey of fat tails in environmental economics. *Annual Review of Resource Economics* 13:289–310.
- Cropper, Maureen L., and Wallace E. Oates. 1992. Environmental economics: A survey. *Journal of Economic Literature* 30 (2): 675–740.
- Cummings, Ronald G., Phillip T. Ganderton, and Thomas McGuckin. 1994. Substitution effects in CVM values. *American Journal of Agricultural Economics* 76 (2): 205–14.
- Darwin, Charles. 1857. Letter no. 2130. *Darwin correspondence project*. <https://www.darwinproject.ac.uk/letter/?docId=letters/DCP-LETT-2130.xml;query=2130;brand=default>.
- Dawson, Deborah, and Jason F. Shogren. 2001. An update on priorities and expenditures under the Endangered Species Act. *Land Economics* 77 (4): 527–32.
- Drupp, Moritz A., and Martin C. Hänsel. 2020. Relative prices and climate policy: How the scarcity of non-market goods drives policy evaluation. *American Economic Journal: Economic Policy* 13 (1): 168–201.
- Duffield, J. 1991. Existence and non-consumptive values for wildlife: Application of wolf recovery in Yellowstone National Park. W-133/Western Regional Science Association Joint Session, Measuring non-market and non-use values, Monterey, CA.
- Faber-Langendoen, D., J. Nichols, L. Master, K. Snow, A. Tomaino, R. Bittman, G. Hammerson, et al. 2012. NatureServe Conservation Status Assessment: Methodology for Assigning Ranks. NatureServe report, Arlington, VA.
- FAO (Food and Agriculture Organization). 2016. <https://www.fao.org/faostat/en/#home>.
- Frank, Eyal. 2020. The impact of the northern spotted owl conservation plan on local labor markets. https://www.eyalfrank.com/wp-content/uploads/2020/02/Frank_NSO_Logging_Jobs.pdf.
- GAO (Government Accountability Office). 2005. Endangered species: Fish and Wildlife Service generally focuses recovery funding on high-priority species, but needs to periodically assess its funding decisions. GAO, Washington, DC.
- Gibbs, Katherine E., and David J. Currie. 2012. Protecting endangered species: Do the main legislative tools work? *PLoS ONE* 7 (5).
- Giraud, K. L., J. B. Loomis, and R. L. Johnson. 1999. Internal and external scope in willingness-to-pay estimates for threatened and endangered wildlife. *Journal of Environmental Management* 56 (3): 221–29.
- Giraud, Kelly, Branka Turcin, John Loomis, and Joseph Cooper. 2002. Economic benefit of the protection program for the Steller sea lion. *Marine Policy* 26 (6): 451–58.
- Google. 2020. Google books Ngram Viewer. <https://storage.googleapis.com/books/ngrams/books/data setsv3.html>.

- Gumbs, Rikki, Claudia L. Gray, Oliver R. Wearn, and Nisha R. Owen. 2018. Tetrapods on the EDGE: Overcoming data limitations to identify phylogenetic conservation priorities. *PLoS ONE* 13 (4): 1–19.
- Haab, Timothy C., Matthew G. Interis, Daniel R. Petrolia, and John C. Whitehead. 2013. From hopeless to curious? Thoughts on Hausman's "dubious to hopeless" critique of contingent valuation. *Applied Economic Perspectives and Policy* 35 (4): 593–612.
- Hagan, Daniel A., James W. Vincent, and Patrick G. Welle. 1992. Benefits of preserving old-growth forests and the spotted owl. *Contemporary Economic Policy* 10 (2): 13–26.
- Hageman, Renatte K. 1985. Valuing marine mammal populations: Benefit valuations in a multi-species ecosystem. National Marine Fisheries Service, Southwest Fisheries Center, Silver Spring, MD.
- Hanemann, W. Michael. 1991. Willingness to pay and willingness to accept: How much can they differ? *American Economic Review* 81 (3): 635–47.
- Hanley, Nick, Robin E. Wright, and Vic Adamowicz. 1998. Using choice experiments to value the environment. *Environmental and Resource Economics* 11 (3–4): 413–28.
- Hartmann, Klaas. 2013. The equivalence of two phylogenetic biodiversity measures: The Shapley value and fair proportion index. *Journal of Mathematical Biology* 67 (5): 1163–70.
- Hausman, Jerry. 2012. Contingent valuation: From dubious to hopeless. *Journal of Economic Perspectives* 26 (4): 43–56.
- Isaac, Nick J. B., David W. Redding, Helen M. Meredith, and Kamran Safi. 2012. Phylogenetically-informed priorities for amphibian conservation. *PLoS ONE* 7 (8): 1–8.
- Jetz, Walter, Gavin H. Thomas, Jeffrey B. Joy, David W. Redding, Klaas Hartmann, and Arne O. Mooers. 2014. Global distribution and conservation of evolutionary distinctness in birds. *Current Biology* 24 (9): 919–30.
- Kerkvliet, Joe, and Christian Langpap. 2007. Learning from endangered and threatened species recovery programs: A case study using U.S. Endangered Species Act recovery scores. *Ecological Economics* 63 (2–3): 499–510.
- King, David A., Deborah J. Flynn, and William W. Shaw. 1988. Total and existence values of a herd of desert bighorn sheep. Western Region Archives 303518, Western Region, Western Extension Directors Association (WEDA), Spokane, WA.
- Kling, Catherine L., Daniel J. Phaneuf, and Jinhua Zhao. 2012. From Exxon to BP: Has some number become better than no number? *Journal of Economic Perspectives* 26 (4): 3–26.
- Kotchen, Matthew J., and Stephen D. Reiling. 2000. Environmental attitudes, motivations, and contingent valuation of nonuse values: A case study involving endangered species. *Ecological Economics* 32 (1): 93–107.
- Krausmann, Fridolin, Karl-Heinz Erb, Simone Gingrich, Helmut Haberl, Alberte Bondeau, Veronika Gaube, Christian Lauk, Christoph Plutzer, and Timothy D. Searchinger. 2013. Global human appropriation of net primary production doubled in the 20th century. *Proceedings of the National Academy of Sciences of the United States of America* 110 (25): 10324–29.
- Lancaster, Tony. 2000. The incidental parameter problem since 1948. *Journal of Econometrics* 95 (2): 391–413.
- Langpap, Christian, and Joe Kerkvliet. 2010. Allocating conservation resources under the Endangered Species Act. *American Journal of Agricultural Economics* 92 (1): 110–24.
- Langpap, Christian, Joe Kerkvliet, and Jason F. Shogren. 2018. The economics of the U.S. Endangered Species Act: A review of recent developments. *Review of Environmental Economics and Policy* 12 (1): 69–91.
- Laurin, Michel. 2010. The subjective nature of Linnean categories and its impact in evolutionary biology and biodiversity studies. *Contributions to Zoology* 4:131–46.

- Lew, Daniel K. 2019. Place of residence and cost attribute non-attendance in a stated preference choice experiment involving a marine endangered species. *Marine Resource Economics* 34 (3): 225–45.
- Lew, Daniel K., David F. Layton, and Robert D. Rowe. 2010. Valuing enhancements to endangered species protection under alternative baseline futures: The case of the Steller sea lion. *Marine Resource Economics* 25 (2): 133–54.
- Lewis, David J., Steven J. Dundas, David M. Kling, Daniel K. Lew, and Sally D. Hacker. 2019. The non-market benefits of early and partial gains in managing threatened salmon. *PlosOne* 14 (8). <https://doi.org/10.1371/journal.pone.0220260>
- Loomis, John B. 1996. Measuring the economic benefits of removing dams and restoring the Elwha River: Results of a contingent valuation survey. *Water Resources Research* 32 (2): 441–47.
- Loomis, John, and Earl Ekstrand. 1997. Economic benefits of critical habitat for the Mexican spotted owl: A scope test using a multiple-bounded contingent valuation survey. *Journal of Agricultural and Resource Economics* 22 (2): 356–66.
- Loomis, John B., and Douglas M. Larson. 1994. Total economic values of increasing gray whale populations: Results from a contingent valuation survey of visitors and households. *Marine Resource Economics* 9 (3): 275–86.
- Melstrom, Richard T. 2021. The effect of land use restrictions protecting endangered species on agricultural land values. *American Journal of Agricultural Economics* 103 (1): 162–84.
- Metrick, Andrew, and Martin L. Weitzman. 1996. Patterns of behavior in endangered species preservation. *Land Economics* 72 (1): 1–16.
- . 1998. Conflicts and choices in biodiversity preservation. *Journal of Economic Perspectives* 12 (3): 21–34.
- Michel, Jean-Baptiste, Yuan Kui Shen, Aviva Presser Aiden, Adrian Veres, Matthew K. Gray, Google Books Team, Joseph P. Pickett, et al. 2011. Quantitative analysis of culture using millions of digitized books. *Science* 331 (6014): 176–82.
- Miller, Julie K., J Michael Scott, Craig R. Miller, and Lisette P. Waits. 2002. The Endangered Species Act: Dollars and sense? *BioScience* 52 (2): 163–68.
- Murray, Brian C., and David N. Wear. 1998. Federal timber restrictions and interregional arbitrage in U.S. lumber. *Land Economics* 74 (1): 76.
- Newbold, Tim, Lawrence N. Hudson, Samantha L. L. Hill, Sara Contu, Igor Lysenko, Rebecca A. Senior, Luca Börger, et al. 2015. Global effects of land use on local terrestrial biodiversity. *Nature* 520 (7545): 45–50.
- Olsen, D., J. Richards, and R. D. Scott. 1991. Existence and sport values for doubling the size of Columbia River basin salmon and steelhead runs. *Rivers* 2 (1): 44–56.
- Pindyck, Robert S. 2013. Climate change policy: What do the models tell us? *Journal of Economic Literature* 51 (3): 860–72.
- Plantinga, Andrew J., Ted L. Helvoigt, and Kirsten Walker. 2014. Critical habitat for threatened and endangered species: How should the economic costs be evaluated? *Journal of Environmental Management* 134:127–35.
- Potter, Kevin M. 2018. Do United States protected areas effectively conserve forest tree rarity and evolutionary distinctiveness? *Biological Conservation* 224:34–46.
- Pounds, J. Alan, Michael P. L. Fogden, and John H. Campbell. 1999. Biological response to climate change on a tropical mountain. *Nature* 398 (6728): 611–15.
- Quigley, John M., and Aaron M. Swoboda. 2007. The urban impacts of the Endangered Species Act: A general equilibrium analysis. *Journal of Urban Economics* 61 (2): 299–318.

- Rubin, J., G. Helfand, and J. Loomis. 1991. A benefit-cost analysis of the northern spotted owl. *Journal of Forestry* 89 (12): 25–30.
- Safi, Kamran, Katrina Armour-Marshall, Jonathan E. M. Baillie, and Nick J. B. Isaac. 2013. Global patterns of evolutionary distinct and globally endangered amphibians and mammals. *PLoS ONE* 8 (5): 4–12.
- Samples, Karl C., and James R. Hollyer. 1990. Contingent valuation of wildlife resources in the presence of substitutes and complements. In *Economic valuation of natural resources: Issues, theory and applications*, 177–92. Abingdon: Routledge.
- Simon, Benjamin M., Craig S. Leff, and Harvey Doerksen. 1995. Allocating scarce resources for endangered species recovery. *Journal of Policy Analysis and Management* 14:415–32.
- Skourtos, M., A. Kontogianni, and P. Harrison. 2010. Reviewing the dynamics of economic values and preferences for ecosystem goods and services. *Biological Conservation* 19:2855–72.
- Solomon, Barry D., Cristi M. Corey-Luse, and Kathleen Halvorsen. 2004. The Florida manatee and ecotourism: Toward a safe minimum standard. *Ecological Economics* 50 (1–2): 101–15.
- Stanley, Denise L. 2005. Local perception of public goods: Recent assessments of willingness-to-pay for endangered species. *Contemporary Economic Policy* 23 (2): 165–79.
- Sterner, Thomas, and U. Martin Persson. 2008. An even Sterner review: Introducing relative prices into the discounting debate. *Review of Environmental Economics and Policy* 2 (1): 61–76.
- Stevens, Thomas H., Jaime Echeverria, Ronald J. Glass, Tim Hager, and Thomas A. More. 1991. Measuring the existence value of wildlife: What do CVM estimates really show? *Land Economics* 67 (4): 390–400.
- Sunding, David, and Jonathan Terhorst. 2014. Conserving endangered species through regulation of urban development: The case of California vernal pools. *Land Economics* 90 (2): 290–305.
- Thomas, Chris D., Alison Cameron, Rhys E. Green, Michel Bakkenes, Linda J. Beaumont, Yvonne C. Collingham, Barend F. N. Erasmus, et al. 2004. Extinction risk from climate change. *Nature* 427:145–48.
- Tunçel, Tuba, and James K. Hammitt. 2014. A new meta-analysis on the WTP/WTa disparity. *Journal of Environmental Economics and Management* 68 (1): 175–87.
- Urban, Mark C. 2015. Accelerating extinction risk from climate change. *Science* 348 (6234): 571–73.
- USFWS (US Fish and Wildlife Service). 2016. Federal and state endangered and threatened species expenditures. USFWS, Washington, D.C.
- Waldron, Anthony, Arne O. Mooers, Daniel C. Miller, Nate Nibbelink, David Redding, Tyler S. Kuhn, J Timmons Roberts, and John L. Gittleman. 2013. Targeting global conservation funding to limit immediate biodiversity declines. *Proceedings of the National Academy of Sciences of the United States of America* 110 (29): 12144–48.
- Wallmo, Kristy, and Daniel K. Lew. 2012. Public willingness to pay for recovering and downlisting threatened and endangered marine species. *Conservation Biology* 26 (5): 830–39.
- . 2016. A comparison of regional and national values for recovering threatened and endangered marine species in the United States. *Journal of Environmental Management* 179:38–46.
- Weitzman, Martin L. 1992. On diversity. *Quarterly Journal of Economics* 107 (2): 363–405.
- . 1993. What to preserve: An application of diversity theory to crane conservation. *Quarterly Journal of Economics* 108 (1): 157–83.
- . 1998. The Noah's Ark problem. *Econometrica* 66 (6): 1279.
- . 2001. Gamma discounting. *American Economic Review* 91 (1): 260–71.
- . 2009. On modelling and interpreting the economics of catastrophic climate change. *Review of Economics and Statistics* 91 (1): 1–19.

- . 2012a. GHG targets as insurance against catastrophic climate damages. *Journal of Public Economic Theory* 14 (2): 221–44.
- . 2012b. What is the “damages function” for global warming: And what difference might it make? *Climate Change Economics* 1:57–69.
- Whitehead, John C. 1992. Ex ante willingness to pay with supply and demand uncertainty: Implications for valuing a sea turtle protection programme. *Applied Economics* 24 (9): 981–88.
- Zoological Society of London. 2008. Edge of Existence Programme. www.edgeofexistence.org.